

PDF Book Store

Website Working Details, Feature Flow, Admin System, Payment Flow, User Tracking aur Deployment Notes

Project Type: Full-stack PDF book selling website

Frontend: React + Vite + Tailwind CSS

Backend: Node.js + Express

Database: MongoDB + Mongoose

Auth: JWT session, password hash, OTP verification

Payment: Razorpay + Manual UPI proof verification

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[Users](#)[Books](#)[Orders](#)[Payments](#)[Visitors](#)[Admin Panel](#)

1. Website Ka Main Purpose

Ye website PDF books aur physical/manual book orders sell karne ke liye bani hai. User account create karta hai, login karta hai, book browse karta hai, payment karta hai, aur successful payment ke baad PDF library ya order history me uska purchase save hota hai. Admin panel se books, orders, users, feedback, payment settings, quotes, support chats aur visitors manage kiye ja sakte hain.

2. Project Structure

Folder/File	Kaam
client	React frontend. User pages aur admin pages yahin hain.
server	Express API, MongoDB models, controllers, routes, auth, payments, uploads.
server/src/models	Database schemas: User, Book, Order, Visitor, Feedback, SupportTicket, PaymentSettings, etc.
server/src/controllers	Business logic: auth, payment, admin, books, visitors, feedback.
server/src/routes	API URL mapping.
uploads	Book covers, PDFs, quotes, payment proof uploads.

3. User Account Flow

1. User signup page par name, email, phone aur password enter karta hai.
2. Password plain text me save nahi hota. Backend bcrypt hash banakar database me save karta hai.
3. Signup ke baad OTP email/phone purpose ke saath generate hota hai.
4. OTP verify hone ke baad user verified ho jata hai.
5. Login password ya OTP flow se hota hai.
6. Login success ke baad JWT token issue hota hai aur active session id save hoti hai.

Login Details Admin Me Kaise Save Hote Hain

Login ya OTP verification success hone par user record me ye fields update hoti hain:

- `firstLoginAt` - pehli baar login ka time.
- `lastLoginAt` - latest login ka time.
- `loginCount` - total successful login count.
- `lastLoginIp` - latest login IP address.
- `lastLoginUserAgent` - browser/device user agent.

Admin panel ke **Manage Users** section me ye login details visible hain.

4. Book Browsing Aur Purchase Flow

1. User home/books page par active books dekhta hai.
2. Book detail page me cover, title, category, price, description aur availability dikhti hai.
3. Digital PDF purchase ke liye user cart/checkout flow use karta hai.
4. Physical/manual book order ke liye delivery details ke saath order book flow use hota hai.

5. Payment Flow

Razorpay Online Payment

1. Checkout par backend Razorpay order create karta hai.
2. User Razorpay checkout me payment karta hai.
3. Payment success ke baad frontend backend ko verification fields bhejta hai.
4. Backend Razorpay signature verify karta hai.
5. Signature valid hone par order status `success` ya manual book ke liye `confirmed` hota hai.
6. Digital PDF order success hone par book user ke `purchasedBooks` me unlock hoti hai.

Manual UPI Payment

1. Agar Razorpay configured nahi hai ya user UPI manual choose karta hai, order pending/submitted state me banta hai.
2. User UPI payment karta hai aur payment screenshot/proof upload karta hai.
3. Admin ko proof admin panel me dikhta hai.
4. Admin status `success` kare to digital PDF unlock hoti hai.
5. Admin physical book order ko `confirmed`, `completed` ya `rejected` kar sakta hai.

Important: User ki order history me ab sirf payment success/confirmed/completed orders dikhte hain. Failed, pending ya submitted orders user history me paid order ke roop me nahi dikhte.

6. User Order History

User dashboard ke **My Orders** page me usi login user ke orders dikhte hain. Backend query user id se filter karti hai, isliye ek user dusre user ka order nahi dekh sakta.

Order Type	User History Me Kab Dikhega
Digital PDF	Payment status <code>success</code> hone ke baad.
Physical/manual book	Status <code>confirmed</code> ya <code>completed</code> hone ke baad.

7. PDF Library Aur Secure PDF Access

PDFs public static file ki tarah directly open nahi hote. User jab library se PDF read/download karta hai, backend JWT token check karta hai aur confirm karta hai ki book user ke purchased list ya successful order me hai.

- Successful digital payment ke baad PDF unlock hoti hai.
- Default digital access duration 30 days hai.
- Admin successful order ke andar PDF access expiry date update kar sakta hai.
- Admin users kisi bhi book ko manage/download kar sakte hain.

8. Admin Panel

Admin route: `/admin/login` . Admin login protected hai aur sirf admin role ke users access kar sakte hain.

Admin Section	Kaam
Dashboard	Total users, books, orders, revenue aur visitors summary.
Add Book	New book add karna, cover/PDF upload karna.
Manage Books	Books edit, activate/deactivate, delete/manage.
Order Book Prices	Physical/manual book prices manage karna.
Manage Orders	Digital aur physical orders, payment proof, status update, access expiry manage.
Support Chats	User support tickets dekhna aur update karna.
User Feedback	User feedback manage karna.
Payment Options	UPI id, payee name, QR image, payment extra charges aur instructions update.
Quote Section	Home/site quote section manage.
Manage Users	All users, phone, role, verification, login count, last login, IP details.
Visitors	Total visits, unique visitors, today visits, page/device/IP/user info.

9. Visitor Tracking Kaise Kaam Karta Hai

1. Frontend me `VisitTracker` component every non-admin route change par run hota hai.
2. Browser localStorage me ek visitor id save hoti hai, jisse same browser ke repeat visits identify hote hain.
3. Frontend `POST /api/site/visit` API call karta hai.
4. Backend visitor details MongoDB me `Visitor` collection me save karta hai.
5. Agar visitor logged-in hai to visitor record me user id bhi attach hoti hai.
6. Admin panel ke Visitors section me latest 300 visit logs aur summary dikhti hai.

Saved visitor data:

- Visitor id
- Logged-in user id, agar available ho
- IP address
- Browser/device user agent
- Visited page
- Referrer
- Screen size
- Language
- Visit time

10. Database Models Summary

Model	Main Fields	Use
User	name, email, phone, password hash, role, verified, login audit, purchasedBooks	User/admin accounts aur login records.
Book	title, description, price, cover, PDF path, category, active status	Book catalog.
Order	user, items, amount, provider, proof, transaction id, status	Payment/order history and unlock logic.
Visitor	visitorId, user, ip, userAgent, page, referrer, screen, language	Website visitor analytics.
PaymentSettings	UPI id, payee name, QR, extra charges, instructions	Payment options admin setting.
Feedback	user/customer feedback and status	Feedback management.
SupportTicket	customer issue/message/status	Support chat/ticket system.

11. Important API Routes

Route	Purpose
POST /api/auth/signup	User signup.
POST /api/auth/verify-otp	Signup OTP verify and login token issue.
POST /api/auth/login	Password login.
POST /api/auth/login/request-otp	Login OTP request.
POST /api/auth/login/verify-otp	OTP login verify.
GET /api/users/orders	Current user's successful order history.
GET /api/users/library	Current user's unlocked PDF library.
POST /api/payments/create-order	Digital checkout order create.
POST /api/payments/verify	Razorpay payment verify.
POST /api/payments/confirm-manual	Manual UPI proof submit.
POST /api/site/visit	Visitor tracking save.
GET /api/admin/stats	Admin dashboard stats.
GET /api/admin/users	Admin users list.
GET /api/admin/orders	Admin orders list.
GET /api/admin/visitors	Admin visitors analytics.

12. Security Points

- Password bcrypt hash ke saath save hota hai.
- JWT secret minimum length required hai.
- Admin routes protected hain: token + role check.
- PDF direct public access nahi hota, backend permission check karta hai.
- Razorpay signature backend par verify hoti hai.
- Payment proof upload admin verification ke liye restricted flow me use hota hai.
- API rate limiters auth/payment/admin write routes par use hote hain.
- Helmet, CORS, compression aur no-store cache headers sensitive auth/admin flows me use hote hain.

13. Deployment Notes

Part	Deployment
Backend	Render. Root directory <code>server</code> , build <code>npm install</code> , start <code>npm start</code> .
Frontend	Vercel/Netlify. Root directory <code>client</code> , build <code>npm run build</code> , publish <code>dist</code> .
Database	MongoDB Atlas via <code>MONGO_URI</code> .
Frontend API URL	<code>VITE_API_URL</code> should point to deployed backend API URL.

14. Environment Variables

- `MONGO_URI` - MongoDB connection string.
- `JWT_SECRET` - secure JWT secret.
- `CLIENT_URL` - allowed frontend URLs.
- `SMTP_USER`, `SMTP_PASS` - email/OTP sending.
- `RAZORPAY_KEY_ID`, `RAZORPAY_KEY_SECRET` - online payment.
- `RAZORPAY_WEBHOOK_SECRET` - webhook verification.
- `UPI_ID`, `UPI_PAYEE_NAME` - manual UPI fallback/defaults.
- `ADMIN_NOTIFY_EMAIL` - payment proof notification email.
- `OWNER_ADMIN_EMAIL` - authorized first admin email.

15. Daily Admin Workflow

1. Admin login kare.
2. Dashboard me users/orders/revenue/visitors check kare.
3. Manage Orders me new payment proof ya book orders verify kare.
4. Successful digital payment ko `success` mark kare, PDF unlock ho jayegi.
5. Physical book order ko `confirmed` ya `completed` kare.
6. Manage Users me user login activity check kare.
7. Visitors section me traffic aur page visits dekhe.
8. Payment Options me UPI/QR/charges update kare agar zarurat ho.

16. Current Local URLs

- Frontend: `http://localhost:5173`
- Backend API: `http://localhost:5000/api`

- Admin Login: <http://localhost:5173/admin/login>
- API Health: <http://localhost:5000/api/health>

Note: Visitor tracking privacy-sensitive hai. Production website par privacy policy me basic analytics/IP/device tracking mention karna recommended hai.